

PROPOSAL DOCUMENTATION

System Architecture Document

Reference architecture for a secure multi-stage application portal supporting applicant intake, reviewer workflows, audit logging, document handling, and POPIA-aligned data protection controls.

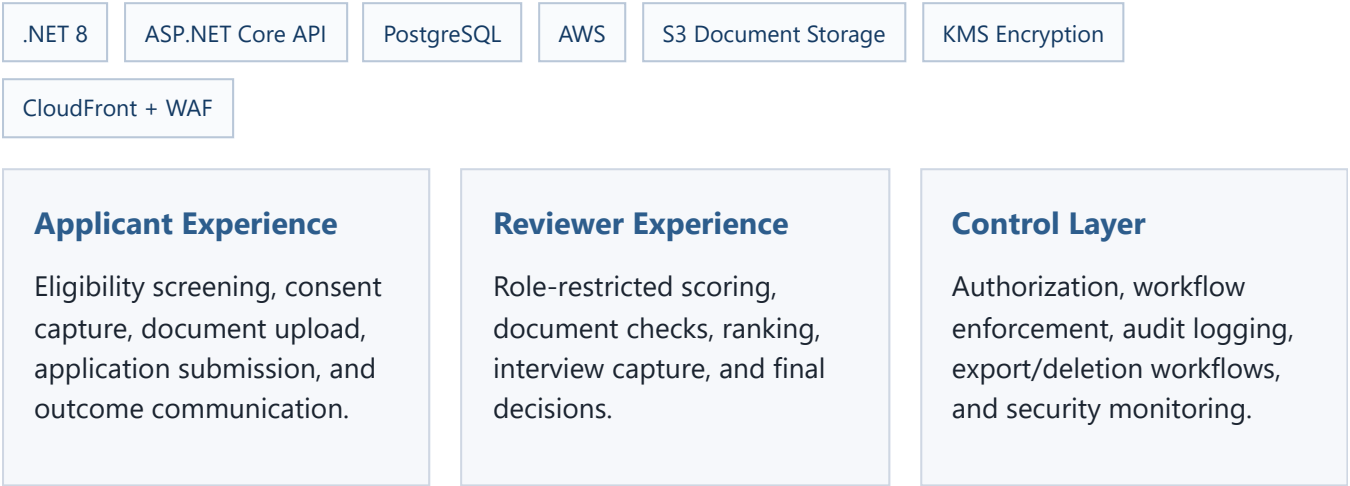
Project	Application Portal
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Prepared to support stakeholder review of the demo portal and the proposed production implementation.

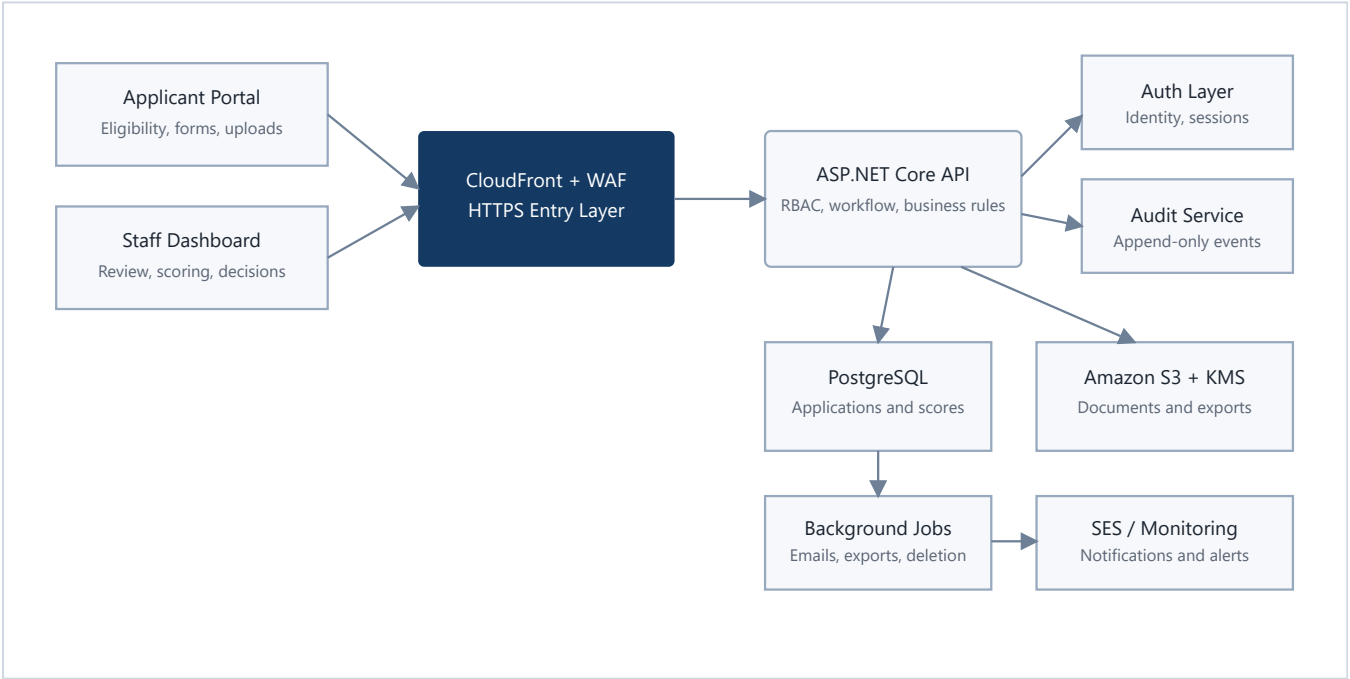
ARCHITECTURE

1. Solution Overview

The proposed solution is a web-based application portal with two primary experiences: an applicant portal for submission and status-related interactions, and a staff dashboard for structured evaluation across multiple stages and roles.



2. High-Level Architecture

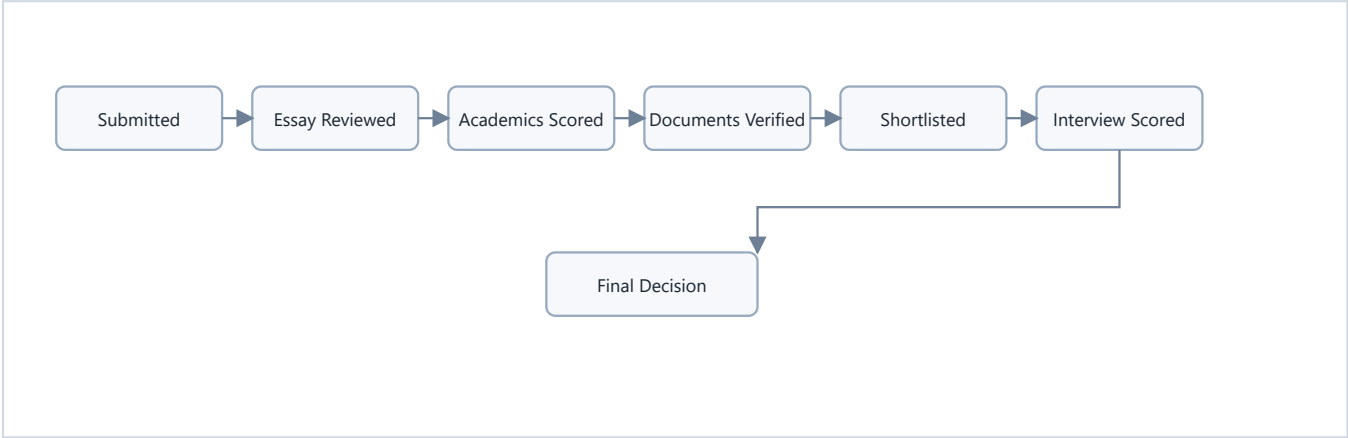


The application is designed around a central API that owns workflow enforcement, authorization, data validation, and audit creation. Client applications never bypass business logic or interact directly with the database or file storage.

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3. Workflow and Access Model

3.1 Workflow State Model



Workflow progression is controlled by explicit transition rules. Users cannot move an application into a later stage while required earlier actions are incomplete. This is particularly important for shortlisting and final decisions, which depend on prior scoring and document verification.

3.2 Role-Based Access Design

Role	Primary Access	Restricted From
Essay Reviewer	Essay responses, basic applicant summary, score entry	Academic marks, peer reviewer scores, final ranking
Academic Reviewer	Transcript capture, subject marks, average calculation	Peer essay scoring, final decisions unless separately authorized
Document Checker	Uploaded files, checklist completion, document issue notes	Reviewer scoring unrelated to document verification
Screener	Combined score views, rank comparison, shortlist decision	Unauthorized editing of prior-stage reviewer records
Interview Admin	Interview scheduling and score sheet capture	Early-stage scoring edits and unrelated user administration
Final Decision Maker	Final decision views, approval/rejection/standby outcomes	Direct modification of preserved audit history

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4. Security and Data Protection Architecture

4.1 POPIA-Oriented Controls

Data in Transit

- TLS 1.2+ across all browser and API traffic
- Secure cookies and session expiry controls
- Signed URLs for sensitive file downloads

Data at Rest

- Encrypted RDS storage
- Encrypted S3 buckets using AWS KMS keys
- Field-level protection for identity numbers and similar fields

Audit and Monitoring

- Append-only audit events for all sensitive actions
- User ID, role, target record, timestamp, and change summary captured
- Monitoring and alerting for suspicious access patterns

Data Subject Rights

- Export package generation
- Cycle-based deletion and retention workflows
- Consent capture and historical retention evidence

4.2 Logical Data Domains

Domain	Examples	Protection Level
Identity Data	Name, contact details, ID/passport data	High
Application Data	Eligibility answers, education, essays, experience	High
Documents	Transcripts, certificates, ID copies, references	High
Reviewer Data	Scores, notes, reviewer allocations	High
System Records	Audit logs, job status, notifications	Medium to High

5. Deployment Recommendation

For a first production deployment, the recommended topology is a static web front end hosted behind CloudFront, a .NET 8 API deployed on App Runner or ECS Fargate, PostgreSQL on Amazon RDS, and encrypted document storage on Amazon S3. This approach supports a clean separation between delivery, application logic, storage, and compliance controls while keeping operations manageable for a small team.

This document is aligned to the current demo scope and the stated production requirements. It is intended for proposal-stage review and can be expanded into a full solution architecture document after approval.